CHAPTER 5 · DERIVATIONS

CIRCULAR MOTION

2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 LINEAR AND ANGULAR SPEED

5.1(C)

RESULT $v = r\omega$

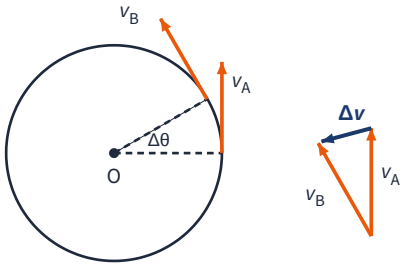
1. Arc length $s = r\theta$ (θ in radians).
2. Differentiate with r constant: $\frac{ds}{dt} = r\frac{d\theta}{dt}$.
3. **$v = r\omega$.**

2 CENTRIPETAL ACCELERATION

5.2(B)

RESULT $a = \frac{v^2}{r} = r\omega^2$

1. An object moves at constant speed v from A to B, turning through a small angle $\Delta\theta$ in time Δt.
2. The velocities v_A and v_B have the same size but different directions. The vector triangle gives the change Δv, which points towards the centre.
3. For small $\Delta\theta$ the triangle is a thin sector: $|\Delta v| = v\Delta\theta$.
4. $a = \frac{\Delta v}{\Delta t} = v\frac{\Delta\theta}{\Delta t} = v\omega$.
5. With $v = r\omega$: **$a = \frac{v^2}{r} = r\omega^2$**, directed towards the centre.



vector triangle: Δv points to O